

Categories, Proofs, and Processes: Assignment 6

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Question 1

Claim 1.1. *h is monotone.*

Proof. Suppose $x, y \in P, x \leq y$. Then

$$\begin{aligned} h(x) &\leq \bigvee \{h(x), h(y)\} \quad (\text{by LUB}) \\ &= h\left(\bigvee \{x, y\}\right) \\ &= h(y) \quad (\text{since } y = \bigvee \{x, y\}) \end{aligned} \tag{1}$$

So h is monotone. □

Proposition 1.2. *U has a left adjoint.*

Proof. Define the functor $F : \mathbf{Set} \rightarrow \mathbf{SL}$

$$F(X) = (\mathcal{P}(X), \subseteq)$$

$$F(f : X \rightarrow Y) : \mathcal{P}(X) \rightarrow \mathcal{P}(Y) := A \mapsto \{f(a) | a \in A\}$$

which is the covariant powerset functor for arrows, so functoriality is satisfied.

Note that $\bigvee S = \bigvee \{A | A \in S\} = \bigvee_{A \in S} A$. The functor on arrows preserves least upper bounds where $\forall f : X \rightarrow Y$ in $\mathbf{Set}$,

$$\begin{aligned} F(f)\left(\bigvee S\right) &= F(f)\left(\bigcup_{A \in S} A\right) \\ &= \left\{f(a) \mid a \in \bigcup_{A \in S} A\right\} \\ &= \bigcup_{A \in S} \{f(a) | a \in A\} \\ &= \bigcup_{A \in S} F(f)(A) \\ &= \bigvee_{A \in S} F(f)(A) \\ &= \bigvee \{F(f)(A) | A \in S\} \end{aligned} \tag{2}$$

Claim 1.3. $F : \mathbf{Set} \rightarrow \mathbf{SL}$ as defined is the left adjoint of U .

F is the left adjoint of U , iff $\forall A \in \text{Ob}(\mathbf{Set})$, there exists a unit $\eta_A : A \rightarrow UFA$ such that

$$\forall f : A \rightarrow B, \exists! \hat{f} : \mathcal{P}(A) \rightarrow B$$

where $f \in \text{Ar}(\mathbf{Set})$, $\hat{f} \in \text{Ar}(\mathbf{SL})$, the following diagram commutes

$$\begin{array}{ccc} A & \xrightarrow{\eta_A} & \mathcal{P}(A) \\ & \searrow f & \downarrow U(\hat{f}) \\ & & B \end{array}$$

For each object $C \in \text{Ob}(\mathbf{Set})$, define the unit arrow $\eta_C : C \rightarrow UFC$ where $UFC = \mathcal{P}(C)$

$$\eta_C(c) = \{c\}$$

From an arrow $f : A \rightarrow B$ in $\mathbf{Set}$, we can construct an arrow in $\mathbf{SL}$

$$\hat{f} : \mathcal{P}(A) \rightarrow B := X \mapsto \bigvee F(f)(X)$$

where $U(\hat{f}) = \hat{f}$. Then $\forall a \in A$,

$$(U(\hat{f}) \circ \eta_A)(a) = \hat{f}(\{a\}) = f(a)$$

So $f = U(\hat{f}) \circ \eta_A$.

Suppose there's another $g : \mathcal{P}(A) \rightarrow B$ such that $f = U(g) \circ \eta_A = g \circ \eta_A$. Then it also preserves least upper bounds where $\forall X \subseteq A$,

$$\begin{aligned} g(X) &= g\left(\bigvee \{\{a\} | a \in X\}\right) \\ &= \bigvee \{g(\{a\}) | a \in X\} \quad (\text{by def. of arrow in SL}) \\ &= \bigvee \{g \circ \eta_A(a) | a \in X\} \quad (\text{since } \eta_A(a) = \{a\}) \\ &= \bigvee \{f(a) | a \in X\} \\ &= \bigvee F(f)(X) \\ &= \hat{f}(X) \end{aligned} \tag{3}$$

so $g = \hat{f}$ is unique. Thus, F is the left adjoint of U . □

Question 2

Proposition 2.4. *The contravariant powerset functor $\mathcal{P} : \mathbf{Set}^{op} \rightarrow \mathbf{Set}$ is its own left adjoint.*

Proof. $\mathcal{P}^{op}$ is the left adjoint of $\mathcal{P}$ iff $\forall X \in \mathbf{Ob}(\mathbf{Set})$, there exists a unit

$$\eta_X : X \rightarrow X$$

such that $\forall f : X \rightarrow \mathcal{P}(Y)$ in $\mathbf{Set}$, $\exists ! \hat{f} : \mathcal{P}^{op}(X) \rightarrow Y$ in $\mathbf{Set}^{op}$ where the diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & X \\ & \searrow f & \downarrow \mathcal{P}(\hat{f}) \\ & & \mathcal{P}(Y) \end{array}$$

Define $\forall X \in \mathbf{Ob}(\mathbf{Set})$, $\eta_X = 1_X$.

Define $\hat{f} = \mathcal{P}^{op}(f)$. Then $\mathcal{P}(\hat{f}) = \mathcal{P}(\mathcal{P}^{op}(f)) = f$ so $f \circ 1_X = f$.

Let $g : \mathcal{P}^{op}(X) \rightarrow Y$ such that $\mathcal{P}(g) \circ 1_X = f$. Then $\mathcal{P}(g) = f$, so $\mathcal{P}^{op}\mathcal{P}(g) = g = \mathcal{P}^{op}(f)$ is unique. □

Question 3

Question 4